

Restaurant Management System

Project Documentation

Project Name : Restaurant Management System

VERCEL LINK : f-hms.vercel.app

<https://f-hms.vercel.app/explore> –customer side link

RENDER LINK : <https://f-hms.onrender.com>

1. Project Overview

The Restaurant Management System is a complete web-based restaurant automation platform developed using the MERN Stack. The application helps restaurants efficiently manage their daily operations including multiple branches, menu management, staff management, inventory, kitchen workflow, billing, reservations, customer management, and reports.

The system provides role-based access so that each user can access only the features assigned to their role.

2. Objectives

- Digitize restaurant operations.
- Reduce manual work.
- Manage multiple restaurant branches.
- Improve order processing.
- Track inventory in real time.
- Simplify billing and payments.
- Generate business reports.
- Provide secure role-based access.

3. Technology Stack

Frontend

- React.js
- Tailwind CSS
- Chart Library

Backend

- Node.js
- Express.js

Database

- MongoDB Atlas

Authentication

- JWT Authentication
- Role-Based Access Control (RBAC)

Image Storage

- Cloudinary

Deployment

- Frontend: Vercel
- Backend: Render

Version Control

- Git & GitHub

4. User Roles

The application supports the following user roles:

- Admin
- Branch Manager
- Chef
- Waiter
- Cashier
- Customer

Each role has dedicated dashboards and permissions.

5. Admin Module

The Admin Dashboard acts as the central control panel.

Features

- Dashboard Overview
- Sales Analytics
- Revenue Reports
- Multi Branch Management
- User Management
- Staff Management
- Role Management
- Menu Management
- Category Management
- Inventory Management
- Supplier Management
- Customer Management
- Reservation Management
- Order Management
- Reports
- Settings

6. Branch Manager Module

The Branch Manager manages operations for a specific branch.

Features

- Branch Dashboard
- Daily Sales
- Staff Management
- Inventory Monitoring
- Menu Availability
- Customer Orders
- Table Reservations
- Reports

7. Chef Module

The Chef Dashboard manages kitchen operations.

Features

- View Incoming Orders
- Accept Orders
- Update Cooking Status
- Mark Orders as Ready
- Kitchen Queue
- Preparation Workflow

Order Status:

Pending → Accepted → Preparing → Ready → Served

8. Waiter Module

Features

- View Tables
- Take Customer Orders
- Update Order Status
- Serve Food
- Customer Requests
- Order History

9. Cashier Module

Features

- Billing
- Invoice Generation
- Payment Collection
- Payment Status
- Order Completion
- Sales Summary

10. Customer Module

Features

- Register/Login
- Browse Menu
- Place Orders

- Reserve Tables
- View Order Status
- Payment
- Order History
- Profile Management

11. Core Modules

Menu Management

- Add Menu
- Edit Menu
- Delete Menu
- Category Management
- Food Images
- Price Management
- Availability Status

Inventory Management

- Raw Material Management
- Stock Updates
- Low Stock Alerts
- Purchase Tracking
- Supplier Mapping

Supplier Management

- Add Suppliers
- Update Supplier Details
- Purchase History
- Contact Management

Reservation Management

- Table Booking
- Reservation Approval
- Reservation Status
- Customer Details

Order Management

- Dine-In Orders
- Takeaway Orders
- Order History
- Order Status Updates

Payment Management

Supported payment methods:

- Cash

Payment Status

- Pending
- Paid
- Failed

12. Dashboard Analytics

The dashboard provides graphical reports including:

- Total Revenue
- Daily Sales
- Monthly Sales
- Orders
- Customers
- Top Selling Items
- Inventory Status
- Branch Performance

13. Authentication Flow

User Login → JWT Token Generated → Role Verification → Dashboard Access → Authorized Operations

14. Database Collections

- Users
- Roles
- Branches
- Categories
- Menu

- Orders
- Reservations
- Customers
- Suppliers
- Inventory
- Payments
- Notifications

15. Project Workflow

Customer Registration/Login → Browse Menu → Place Order → Waiter Receives Order → Chef Accepts Order → Food Preparation → Ready for Serving → Cashier Generates Bill → Payment Completed → Order Closed

16. Security Features

- JWT Authentication
- Password Encryption
- Role-Based Authorization
- Input Validation
- Secure API Endpoints

17. Key Features

- Multi-Branch Support
- Responsive Design
- Role-Based Dashboards
- Real-Time Order Workflow
- Inventory Tracking
- Reservation System
- Billing System
- Reports & Analytics
- Cloud Image Upload
- Secure Authentication

19. Project Outcome

The Advanced Restaurant Management System successfully automates restaurant operations through a centralized platform. The application improves operational efficiency, minimizes manual work, enables better inventory control, streamlines order processing, enhances

customer experience, and provides real-time business insights through dashboards and analytics.

Conclusion

This project demonstrates a scalable restaurant management solution built using the MERN Stack with modern architecture, secure authentication, role-based access control, cloud image storage, and deployment on cloud platforms. The system is designed to support restaurants of various sizes and can be extended into a SaaS platform for multiple restaurant businesses.